



Numerical analysis of hydrogen absorption in a large-scale modular metal hydride reactor with a hybrid heat exchanger comprising an internal finned tube and an external helical coil

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ARTICLE INFO

Keywords:

Metal hydride reactor
Thermal management
Hydrogen absorption
Numerical simulation
Dual-circuit heat exchanger

ABSTRACT

Effective thermal management is a critical bottleneck for large-scale metal hydride (MH) hydrogen storage systems, demanding innovative heat exchanger (HX) designs. This study numerically investigates a novel, modular 400 kg-class Metal Hydride Reactor (MHR). The reactor features a hybrid dual-circuit HX that combines an internal finned tube with an external helical coil. Our analysis is based on a multi-physics model validated against experimental data. The analysis reveals a performance bottleneck that shifts from external convection to internal conduction; increasing the heat transfer fluid (HTF) velocity reduces the peak bed temperature by 15.7 K, but demonstrates clear diminishing returns. A comparative analysis confirms the synergistic advantage of the dual-circuit design, which significantly outperforms configurations using only internal or external cooling. Increasing the bed's solid-phase thermal conductivity from 1.6 to 9.6 W/(m K) is shown to shorten the 90% absorption time by approximately 24%. Furthermore, the hydrogen absorption rate is found to be predominantly governed by the thermodynamic driving force; increasing the supply pressure from 5 to 9 MPa halves the 90% absorption time to 3600 s. These findings establish a quantitative framework for co-optimizing system components, paving the way for efficient and scalable next-generation hydrogen storage systems based on this modular reactor design.

1. Introduction

The escalating global climate crisis and the global pursuit of carbon neutrality have catalyzed a profound transition in the world's energy systems. A central tenet of this transformation is the replacement of fossil fuels with clean and sustainable alternatives. Hydrogen has emerged as a key component in a future sustainable energy mix, owing to its high specific energy density, zero-emission nature, and versatility as an energy carrier [1,2]. However, the transition to a hydrogen economy requires overcoming significant technological and economic challenges, particularly the development of safe, dense, and cost-effective hydrogen storage technologies. Conventional storage methods, such as high-pressure gas compression and cryogenic liquefaction, are technologically mature but constrained by low volumetric density, high energy penalties, and significant safety concerns, limiting their widespread

application [3,4]. In contrast, solid-state storage in metal hydrides (MH) presents a compelling alternative. MH systems reversibly store hydrogen in a solid matrix. This approach allows them to achieve high volumetric densities while operating at much lower pressures, which offers inherent safety advantages [5,6]. These attributes make MH technology particularly attractive for applications where space and safety are paramount, motivating decades of research into advanced materials and reactor designs [7,8].

The practical performance of a metal hydride reactor (MHR) is determined not only by the intrinsic properties of the storage material but also by the engineering of the reactor system itself [9,10]. The hydrogen absorption and desorption cycles are strongly exothermic and endothermic, respectively, creating a major thermal management challenge. This challenge is exacerbated by the intrinsically poor thermal

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<https://doi.org/10.1016/j.ijheatmasstransfer.2025.127987>

Received 13 August 2025; Received in revised form 30 September 2025; Accepted 16 October 2025

Available online 22 October 2025

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conductivity of the MH powder bed, which impedes heat transport to and from the reaction zones. As a result, large thermal gradients form within the reactor, and the overall reaction rate becomes limited by heat transfer efficiency rather than by the material's intrinsic kinetics [11]. Multi-physics simulations confirm this behavior, illustrating slow-moving reaction fronts that originate from the cooled walls and stall before reaching the reactor core [12,13]. Therefore, realizing the full potential of advanced MH materials requires an effective solution to this fundamental heat transfer problem through innovative reactor design.

To overcome this heat transfer bottleneck, research efforts have primarily focused on two strategies: enhancing the bed's effective thermal conductivity [14,15], and optimizing the reactor's heat exchanger (HX) architecture [16]. The former often involves incorporating high-conductivity additives into the MH powder. Kim et al. [17] recently demonstrated that adding just 2 wt% of reduced graphene oxide (rGO) to create an rGO/Mg nanocomposite packed bed could increase the effective thermal conductivity by at least 25% compared to bulk MgH_2 . Zhan et al. [18] demonstrated that pressing the alloy powder with graphite and a binder into compacted discs increased the bed's effective thermal conductivity by 5–8 times, which in turn shortened the total reaction time by nearly 50% compared to a simple loose powder bed. However, the most common approach is to enhance the HX architecture. Early efforts to improve heat exchange included the addition of internal active components, such as heat exchange fluid pipes [19,20], or passive components like high thermal conductivity fins [21,22]. More advanced designs have explored complex internal architectures, including helical coils [23,24] and annular configurations [25,26], which significantly improve thermal uniformity. Wang et al. [27] used entransy theory to redesign the fin profile. Their work showed that an optimized fin shape could shorten the characteristic hydrogenation time by 23% compared to a conventional uniform thickness fin. Jiang et al. [28] found that a TPMS-based reactor could achieve a final hydrogen absorption reaction fraction of 0.926, a dramatic improvement over the 0.642 reached in a traditional finned reactor under identical conditions. Yoo et al. [26] numerically investigated a thin double-layered annular ZrCo bed that demonstrated extremely rapid kinetics, achieving 99% hydrogen absorption in approximately 14 min. However, these approaches introduce manufacturing complexity and critical safety risks associated with potential coolant leakage into the high-pressure hydrogen environment. External HXs separate the heat transfer fluid (HTF) from the MH bed. The efficacy of purely external systems, however, is fundamentally constrained by the high thermal resistance of the reactor wall and the low-conductivity MH bed, resulting in the formation of hot spots in the reactor core. This intrinsic limitation has driven the adoption of hybrid heat exchange solutions. A typical configuration combines internal finned tubes with an external jacketed shell. This approach is exemplified in the shell-and-tube design simulated by Afzal et al. [29], where the shell acts as an external jacket for multiple internally-finned MH tubes, and in the annular finned reactors developed by Parashar et al. [30] and Prasad & Muthukumar [31], which also utilize an external jacket for cooling the periphery. However, the conventional jacketed shell configuration is often characterized by large HTF inventory and poor convective heat transfer. As a result, reactor design therefore faces the persistent challenge of synergizing a safe internal structure with a more efficient external cooling technology to overcome the sluggishness of conventional jackets.

Significant progress has been made in the development of large-scale MHR systems [32,33]. Landmark achievements include the work of Kubo et al. who designed a MH tank capable of storing up to 1000 N m³ of hydrogen using a combination of an external water-cooling jacket and internal fins [34]. On an even larger scale, Ding's team developed a ton-class magnesium-based hydrogen storage vehicle, which relies on internal U-type tubes for heat management [35]. However, evaluating novel HX architectures through physical prototyping

is often expensive and complex for large-scale MHRs. Consequently, numerical simulation has become an indispensable tool for virtual prototyping, parametric analysis, and system optimization [36,37]. Multiphysics models, which couple reaction kinetics with conjugate heat transfer and fluid dynamics, offer precise spatiotemporal resolution of temperature, pressure, and concentration fields that cannot be captured by discrete experimental measurements. However, while numerous simulation studies have explored individual internal or external cooling strategies, thorough numerical investigations of advanced hybrid HX systems designed for enhanced thermal management are notably lacking. A rigorous analysis is therefore required to validate the potential of such innovative designs and to elucidate their underlying thermo-kinetic behavior.

This study introduces and numerically evaluates a novel hybrid modular thermal management system for large-scale MHRs. The fundamental unit of the proposed system is a cylindrical reactor featuring two distinct, synergistic active cooling circuits. The first is an external helical half-pipe coil, a design chosen for its high efficiency and reduced required HTF volume. This efficiency stems from induced secondary flows (i.e., Dean vortices) that disrupt the thermal boundary layer. This enhancement is significant; for instance, Mahir et al. [38] demonstrated that under identical Reynolds number conditions, the Nusselt number for a helical coil is nearly four times higher than that of a conventional wall jacket. Furthermore, Verma et al. [39] showed that inducing swirl flow can increase the Nusselt number by up to 2.6 times compared to smooth, straight tube flow. This external coil is synergistically combined with a second internal circuit: a single central tube equipped with six longitudinal fins. This internal circuit is designed to directly extract heat from the reactor core while minimizing structural complexity. This dual-circuit architecture forms the basis of the modular design, allowing for scalable hydrogen capacity by paralleling multiple reactor units. An experimentally validated, comprehensive, three-dimensional multi-physics model is developed to investigate the system's thermo-kinetic performance across a range of operational parameters. The core objectives are to elucidate the governing heat transfer mechanisms and to quantify the effectiveness of the dual-circuit cooling design in mitigating thermal gradients and enhancing hydrogen absorption rates. Ultimately, this work provides a validated design framework and delivers critical insights into the competing heat transfer limitations, guiding the optimization of scalable and modular solid-state hydrogen storage systems.

2. Reactor design and experimental setup

The experimental apparatus is detailed in Fig. 1, which provides a photograph of the complete system (a), a 3D schematic of the reactor module (b), and its corresponding 2D engineering drawings (c). The system consists of a MHR module, a hydrogen supply unit, a thermal management unit, and a data acquisition and control unit. The MHR module features a modular, dual-vessel architecture, with each cylindrical pressure vessel having an overall length of 2400 mm and an outer diameter of 325 mm, as detailed in Fig. 1(b). The vessel body and heat exchange tubes are constructed from 316L stainless steel, with high-conductivity aluminum alloy used for the fins. A key feature is the dual-circuit active heat exchange system. The internal HX consists of a single, continuous central tube with an outer diameter (OD) of 50 mm and an inner diameter (ID) of 30 mm. Welded to this central tube are six longitudinal, plate-shaped aluminum fins, each with a radial length of 100 mm. The external HX is a helical half-pipe coil with a 38 mm OD and 30 mm ID, which is welded onto the vessel's exterior and completes 16.5 turns with an axial pitch of 100 mm. The inlets and outlets for the internal and external HTF circuits and the hydrogen port are located on the end flanges. For both circuits, the HTF flow is arranged in a co-current configuration. Four K-type thermocouples (TC1–TC4) were installed in each vessel to monitor the spatio-temporal temperature distribution as shown in Fig. 1(c). The thermocouples were placed at

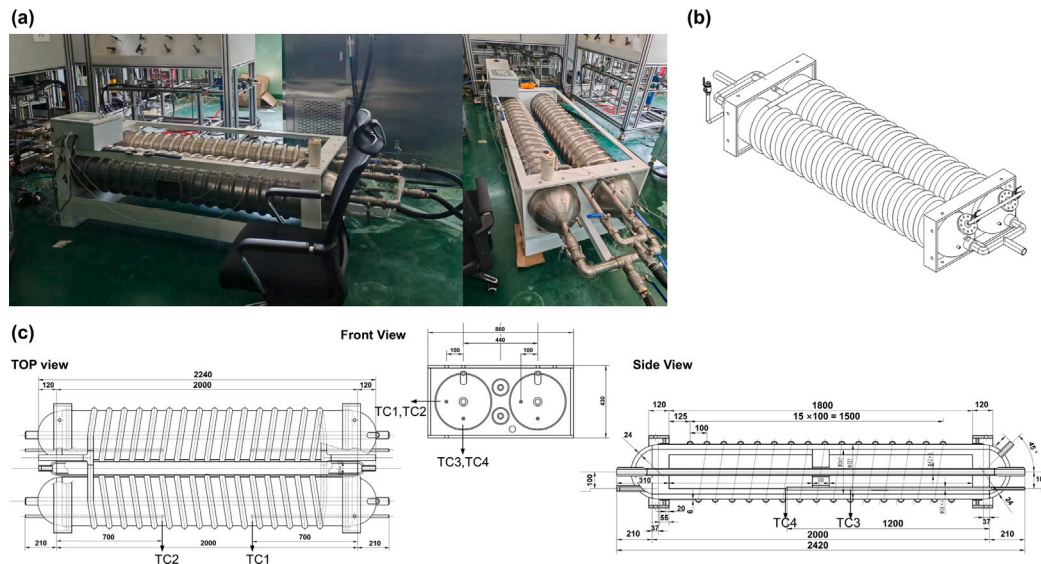


Fig. 1. The MHR experimental system. (a) Photograph of the complete experimental apparatus, including the MHR module, control cabinet, and thermal management unit. (b) 3D schematic of the dual-vessel MHR module. (c) 2D engineering drawings of the MHR module showing key dimensions and cross-sectional views.

two distinct axial cross-sections: TC1 and TC3 were located 700 mm from the end flange, while TC2 and TC4 were at 1300 mm. At these locations, their radial and angular positions were chosen to differentiate between regions with and without direct fin influence. Specifically, TC1 and TC2 were positioned at the side of the vessel, in the bulk powder region away from the fins. In contrast, TC3 and TC4 were placed at the bottom of the vessel, in close proximity to the internal fins to capture their direct thermal impact. This dual-circuit hybrid configuration was designed to address the competing demands of safety and performance. The external helical coil offers high heat transfer efficiency with a reduced coolant volume compared to a conventional jacket. This is synergistically combined with the internal finned tube, which directly targets the reactor core where hotspots are most likely to form, thus ensuring more uniform thermal management throughout the large-scale bed.

The hydrogen storage material used was a commercial Ti-Mn based AB_2 -type Laves phase alloy, $Ti_{1-x}Zr_x(Cr-Mn-Fe)_2$, supplied by Zhongke Xuanda New Energy Technology Inc. Each reactor vessel was filled with 400 kg of the alloy. The fundamental thermodynamic properties of the alloy were first characterized by measuring its Pressure–Composition–Temperature (PCT) isotherms using a Sieverts-type apparatus (GRIMAT-PCT-SD) with high-purity (99.99%) hydrogen. Prior to performance evaluation in the large-scale MHR, the as-received material underwent an in-situ activation procedure. This procedure began with evacuating the loaded reactor to approximately 10 Pa, followed by heating the material to 70 °C for 6 h under vacuum. After the reactor was cooled to ambient temperature, it was pressurized with high-purity hydrogen to 6 MPa. Finally, two complete absorption-desorption cycles were performed to ensure full activation. The thermal management system was composed of a 2 N m³ thermostatic circulation unit and a data acquisition module. In the system, the HTF was circulated by a horizontal single-stage centrifugal pump, while an integrated self-circulating pump within the thermostatic unit ensured uniform HTF temperature and effectively reduced fluctuations in the outlet temperature. An ASAIR AFD4100A33 vortex flowmeter was equipped to monitor the flow rate. For monitoring the state of the hydrogen storage system, WZP-035 precision platinum resistance temperature sensors with an accuracy of ± 0.15 K and SUO-PX300 model pressure transmitters from Hangzhou Meiyi were used. For data acquisition, a KSB12A6R-1C paperless recorder from Ningbo Kesun Instrument was selected to precisely collect temperature, pressure, and flow rate data,

providing fully isolated, high-precision sampling. Following activation, the MHR's performance was evaluated through a series of absorption and desorption experiments. For the specific validation experiment, the HTF was first circulated through both heat exchange circuits at a constant inlet temperature of 298.15 K and a flow rate corresponding to 0.4 m/s until all four thermocouples (TC1–TC4) showed stable readings within ± 0.2 K of the HTF temperature for at least 30 min. The absorption process was then initiated by supplying high-purity hydrogen at a pressure of 7 MPa to the reactor. Throughout the test, temperature profiles from thermocouples TC1–TC4 and the system pressure were continuously recorded until the reaction reached near completion.

3. Numerical model

To investigate the complex thermo-kinetic behavior of the novel dual-circuit MHR during hydrogen absorption, a comprehensive three-dimensional numerical model was developed. The model is based on a simplified geometry, representing a single reactor from the dual-vessel experimental module to ensure the problem is computationally tractable. Fig. 2(a) illustrates this geometric simplification and Fig. 2(b) delineates the distinct computational domains used in the simulation. As shown in Fig. 2(b), the model is partitioned into several key domains: the porous MH bed, the solid vessel components, the internal and external HTF circuits, and the thin-walled fins. The model, implemented on the COMSOL Multiphysics platform, uses the finite element method (FEM) to solve the fully coupled equations governing reaction kinetics, heat transfer, and mass transport within these domains.

3.1. Kinetic and thermodynamic equations

The hydrogen absorption process within a MH bed is a complex gas–solid reaction. Its reaction rate can be described by the Johnson–Mehl–Avrami–Kolmogorov (JMAK) model [40–42], where the rate of change in hydrogen concentration, c (wt%), is expressed as:

$$\frac{dc}{dt} = A \exp\left(-\frac{E_a}{RT}\right) \ln\left(\frac{P}{P_{eq}}\right) (c_{max} - c) \quad (1)$$

where A is the pre-exponential factor (s^{−1}), E_a is the apparent activation energy (J/mol), R is the universal gas constant (8.314 J mol^{−1} K^{−1}), and T is the local temperature (K). The driving force is represented by the pressure ratio, and the term $(c_{max} - c)$ accounts for

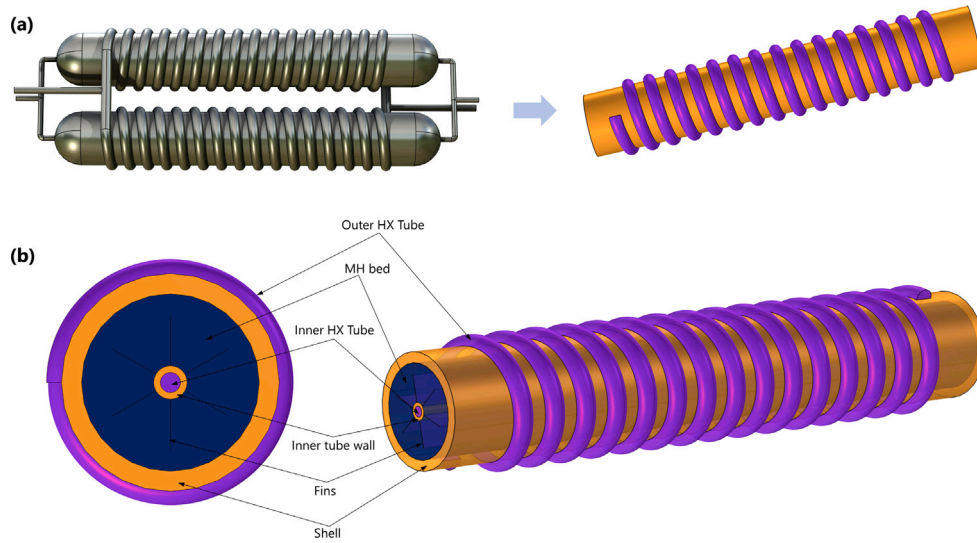


Fig. 2. Schematic of the computational model and domains. (a) Geometric simplification from the dual-vessel experimental module to the single-vessel computational model. (b) Cross-sectional and 3D views of the computational domains, identifying the MH bed, Inner HX Tube, Outer HX Tube, Fins, Inner tube wall, and Shell.

the diminishing reaction potential as the bed approaches its maximum storage capacity, c_{max} (1.88 wt%).

The equilibrium pressure P_{eq} is dependent on both temperature (T) and hydrogen concentration (c). This relationship is modeled using the Van't Hoff equation, which correlates the equilibrium pressure at a given temperature to a reference PCT isotherm:

$$P_{eq}(T, c) = P_{iso}(T_{ref}, c) \exp \left(\frac{\Delta H}{R} \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) \right) \quad (2)$$

where ΔH is the enthalpy of absorption (J/mol), and T_{ref} is the reference temperature (293.15 K). The reference isotherm $P_{iso}(T_{ref}, c)$ is implemented as a cubic spline interpolation of the measured PCT data.

3.2. Governing equations for mass and momentum transfer

The transport of hydrogen gas through the porous MH bed is governed by the principles of mass conservation and Darcy's Law. The continuity equation, which accounts for the consumption of hydrogen during the absorption reaction, is given by:

$$\frac{\partial(\delta \rho_g)}{\partial t} + \nabla \cdot (\rho_g \mathbf{u}) = Q_m \quad (3)$$

where δ is the porosity of the bed, ρ_g is the density of hydrogen gas, and $\mathbf{u}$ is the superficial velocity vector of the gas. The mass sink term, Q_m ($\text{kg m}^{-3} \text{s}^{-1}$), represents the rate of hydrogen consumption by the alloy and is directly linked to the reaction kinetics:

$$Q_m = -(1 - \delta) \rho_s \frac{dc}{dt} \quad (4)$$

here, ρ_s (kg m^{-3}) is the density of the solid MH alloy.

The gas velocity $\mathbf{u}$ is determined by Darcy's Law, which relates the fluid velocity to the pressure gradient within the porous medium:

$$\mathbf{u} = -\frac{K}{\mu_g} \nabla P \quad (5)$$

where K is the intrinsic permeability of the porous bed (m^2) and μ_g is the dynamic viscosity of hydrogen gas (Pa s).

The permeability K is given by:

$$K = \frac{\delta^3 d_p^2}{150(1 - \delta)^2} \quad (6)$$

where d_p is the mean particle diameter of the MH powder.

3.3. Governing equation for heat transfer

The model accounts for heat generation from the reaction, conduction through solids and the porous bed, and convection within the HTF circuits. A single energy equation is used, assuming local thermal equilibrium (LTE) between the gas and solid phases in the MH bed. A detailed physical justification for this assumption is provided in the Supplementary Material S1.

In the porous MH bed:

$$(\rho C_p)_e \frac{\partial T}{\partial t} + \rho_g C_{p,g} (\mathbf{u} \cdot \nabla T) = \nabla \cdot (\lambda_e \nabla T) + Q_T \quad (7)$$

where the effective volumetric heat capacity, $(\rho C_p)_e$ ($\text{J m}^{-3} \text{K}^{-1}$), and the effective thermal conductivity, λ_e ($\text{W m}^{-1} \text{K}^{-1}$), are volume-averaged properties of the solid and gas phases. These are defined as:

$$(\rho C_p)_e = \delta \rho_g C_{p,g} + (1 - \delta) \rho_s C_{p,s} \quad (8)$$

$$\lambda_e = \delta \lambda_g + (1 - \delta) \lambda_s \quad (9)$$

Here, δ is the bed porosity, ρ is the density (kg m^{-3}), C_p is the specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$), and λ is the thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$), with subscripts s and g denoting the solid alloy and hydrogen gas, respectively. The volumetric heat source Q_T (W m^{-3}) from the exothermic reaction is given by:

$$Q_T = -(1 - \delta) \rho_s \frac{dc}{dt} \frac{\Delta H}{M_{H_2}} \quad (10)$$

where M_{H_2} is the molar mass of hydrogen.

For the solid components, including the container walls and the HX tubes, heat transfer is governed by the standard heat conduction equation:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (\lambda \nabla T) \quad (11)$$

For the thin aluminum fins, a Thin Layer model is employed to provide a computationally efficient representation. It simplifies the 3D problem into a 2D problem solved on the boundary surfaces of the fins. The governing equation is:

$$d_s \rho_{fin} C_{p,fin} \frac{\partial T}{\partial t} + \nabla_t \cdot \mathbf{q}_t = q_{net} \quad (12)$$

where d_s is the fin thickness (m), ρ_{fin} is the fin density (kg m^{-3}), and $C_{p,fin}$ is its specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$). The term on the

left represents the transient storage of thermal energy within the fin material. The second term, $\nabla_i \cdot \mathbf{q}_i$, represents the divergence of the heat flux conducted tangentially along the fin's surface, where the tangential heat flux vector $\mathbf{q}_i$ is given by:

$$\mathbf{q}_i = -d_s \lambda_{fin} \nabla_i T \quad (13)$$

Here, λ_{fin} is the thermal conductivity of the fin, and ∇_i is the surface gradient operator. The term on the right, q_{net} , represents the net heat flux per unit area transferred to the fin from the surrounding medium across its top and bottom surfaces.

3.4. Governing equations for the HTF

The flow of the HTF in the internal and external HX tubes is modeled using a full three-dimensional computational fluid dynamics approach. The flow is assumed to be turbulent and is described by the Reynolds-Averaged Navier–Stokes (RANS) equations. The standard k - ϵ turbulence model is used to account for turbulent effects on momentum and heat transport. The governing equations for the incompressible HTF are:

$$\nabla \cdot (\rho_f \mathbf{u}_f) = 0 \quad (14)$$

where ρ_f is the density of the HTF and $\mathbf{u}_f$ is the time-averaged velocity vector (m s^{-1}).

The conservation of momentum is described by the RANS equation:

$$\rho_f \frac{\partial \mathbf{u}_f}{\partial t} + \rho_f (\mathbf{u}_f \cdot \nabla) \mathbf{u}_f = -\nabla p_f + \nabla \cdot \boldsymbol{\tau}_f + \nabla \cdot \boldsymbol{\tau}_T \quad (15)$$

where p_f is the time-averaged pressure. The term $\boldsymbol{\tau}_f$ represents the viscous stress tensor for the mean flow:

$$\boldsymbol{\tau}_f = \mu_f (\nabla \mathbf{u}_f + (\nabla \mathbf{u}_f)^T) \quad (16)$$

where μ_f is the dynamic viscosity of the HTF. The term $\boldsymbol{\tau}_T$ is the Reynolds stress tensor, which represents the additional momentum transport due to turbulent fluctuations.

To model the Reynolds stress tensor and close the RANS equations, this study utilizes the standard k - ϵ turbulence model. This two-equation model is based on the Boussinesq hypothesis, which relates the Reynolds stresses to the mean velocity gradients via a turbulent viscosity, μ_T :

$$\boldsymbol{\tau}_T = \mu_T (\nabla \mathbf{u}_f + (\nabla \mathbf{u}_f)^T) - \frac{2}{3} \rho_f k \mathbf{I} \quad (17)$$

where k is the turbulent kinetic energy ($\text{m}^2 \text{s}^{-2}$) and $\mathbf{I}$ is the identity matrix. The turbulent viscosity μ_T is calculated from two additional transport equations for k and its dissipation rate, ϵ ($\text{m}^2 \text{s}^{-3}$).

The conservation of energy within the HTF is governed by the following equation, which includes convective transport and diffusive heat flux from both molecular and turbulent effects:

$$\rho_f C_{p,f} \frac{\partial T_f}{\partial t} + \rho_f C_{p,f} (\mathbf{u}_f \cdot \nabla T_f) = \nabla \cdot [(\lambda_f + \lambda_T) \nabla T_f] + \Phi_v \quad (18)$$

where T_f is the mean fluid temperature, $C_{p,f}$ is its specific heat capacity, and the term Φ_v represents heat generation from viscous dissipation. The diffusion term, $\nabla \cdot [(\lambda_f + \lambda_T) \nabla T_f]$, models heat flux through two distinct mechanisms. The fluid's molecular thermal conductivity, λ_f , is a material property that governs heat transfer via molecular collisions and is dominant in the near-wall viscous sublayer. In contrast, the turbulent thermal conductivity, λ_T , is a flow characteristic that models the significantly enhanced heat transport resulting from vigorous eddy mixing in the bulk turbulent flow.

The turbulent thermal conductivity is determined from the turbulent viscosity μ_T via the turbulent Prandtl number, Pr_T :

$$\lambda_T = \frac{\mu_T C_{p,f}}{Pr_T} \quad (19)$$

The Kays–Crawford model [43] is used to dynamically calculate the local Pr_T . To accurately resolve the steep near-wall gradients without prohibitive computational cost, a thermal wall function approach is utilized.

3.5. Initial and boundary conditions

To solve the equations, appropriate initial and boundary conditions are specified.

Initial Conditions ($t = 0$): the MHR is assumed to be in a state of thermal and chemical equilibrium. The initial temperature throughout the reactor bed, cooling tube walls, and fins is uniform.

$$T(t = 0) = T_0 \quad (\text{All domains}) \quad (20)$$

$$P(t = 0) = P_0 \quad (\text{MH Bed}) \quad (21)$$

$$c(t = 0) = c_0 \quad (\text{MH Bed}) \quad (22)$$

$$T_f(z, t = 0) = T_{f,0} \quad (\text{HTF Circuits}) \quad (23)$$

where T_0 , P_0 , c_0 , and $T_{f,0}$ are the prescribed initial temperature, pressure, reaction extent, and HTF temperature, respectively. Typically, P_0 is set to atmospheric pressure, and c_0 is set to 0 for an absorption process.

Boundary Conditions ($t > 0$): A time-dependent pressure condition was imposed at the hydrogen inlet to simulate gradual pressurization. A linear ramp function was used to increase the pressure from P_{atm} to the final absorption pressure, P_{abs} , over a time interval t_{ramp} of 3600 s:

$$P(t)|_{\Gamma_{in}} = \begin{cases} P_{atm} + (P_{abs} - P_{atm}) \frac{t}{t_{ramp}} & \text{for } 0 \leq t \leq t_{ramp} \\ P_{abs} & \text{for } t > t_{ramp} \end{cases} \quad (24)$$

All external surfaces of the reactor module exposed to the ambient environment were modeled with a convective heat transfer boundary condition to account for natural convection heat loss:

$$-\lambda \frac{\partial T}{\partial \mathbf{n}} = H_e (T_{wall} - T_{amb}) \quad (25)$$

At the inlets of both the internal and external HTF circuits, a uniform velocity profile ($\mathbf{u}_{inlet}$) and a constant temperature (T_{inlet}) were prescribed. At the outlets, a zero relative pressure condition ($p_f = 0$) was set to allow for free outflow.

At all fluid–solid interfaces, a no-slip boundary condition for velocity was applied. Thermal continuity was enforced across all internal solid–solid and solid–fluid boundaries. At the welded solid–solid interfaces, the thermal contact resistance was considered negligible. This simplification is justified by a detailed order-of-magnitude analysis, which confirms its contribution is several orders of magnitude smaller than that of the bed's conduction resistance (see Supplementary Material S2).

3.6. Numerical implementation

The fully coupled system of partial differential equations described above was discretized in space using the Finite Element Method. A transient study was performed to capture the evolution of pressure, temperature, and hydrogen concentration over time. The time-dependent problem was solved using a backward differentiation formula (BDF) time-stepping scheme with adaptive step-size control. The resulting system of nonlinear algebraic equations at each time step was solved using a fully coupled solver employing a direct solver (PARDISO) for the linear sub-problem. A detailed description of the meshing strategy and mesh independence study is provided in Supplementary Material S3. The thermophysical properties for all materials used in the numerical model are summarized in Table 1. Detailed Physics Interfaces and solver configurations are provided in the Supplementary Material S4 and S5.

Table 1

Representative thermophysical parameters used in the simulations at $T = 293.15$ K.

Component	Parameter	Symbol	Value
Metal hydride bed			
	Density	ρ_s	6800 kg/m ³
	Porosity	δ	0.5
	Specific heat capacity	$C_{p,s}$	480 J/(kg K)
	Thermal conductivity	λ_s	1.6 W/(m K)
	Mean particle diameter	d_p	3×10^{-4} m
Hydrogen gas			
	Density	ρ_g	0.0899 kg/m ³
	Viscosity	μ_g	8.67×10^{-6} Pa s
	Specific heat capacity	$C_{p,g}$	14287 J/(kg K)
	Thermal conductivity	λ_g	0.1769 W/(m K)
Heat transfer fluid (Water)			
	Density	ρ_f	998.1 kg/m ³
	Viscosity	μ_f	0.001 Pa s
	Specific heat capacity	$C_{p,f}$	4180 J/(kg K)
	Thermal conductivity	λ_f	0.597 W/(m K)
Heat exchanger fins (Aluminum)			
	Density	ρ_{fin}	2700 kg/m ³
	Specific heat capacity	$C_{p,fin}$	903 J/(kg K)
	Thermal conductivity	λ_{fin}	236 W/(m K)
Heat exchanger tubes (316L Steel)			
	Density	ρ_{tu}	8000 kg/m ³
	Specific heat capacity	$C_{p,tu}$	467 J/(kg K)
	Thermal conductivity	λ_{tu}	14.6 W/(m K)

Table 2

Geometric parameters and baseline operating conditions for the numerical model.

Parameter	Symbol	Value	Unit
Geometric parameters			
Vessel overall length	L_{ves}	2400	mm
Vessel outer diameter	D_{ves}	325	mm
Inner HX tube outer diameter	$D_{in,o}$	50	mm
Inner HX tube inner diameter	$D_{in,i}$	30	mm
Number of fins	N_{fin}	6	–
Fin radial length	L_{fin}	100	mm
Fin thickness	d_f	2	mm
External coil outer diameter	$D_{coll,o}$	38	mm
External coil inner diameter	$D_{coll,i}$	30	mm
Baseline operating conditions			
Initial temperature (all domains)	T_0	298.15	K
Initial bed pressure	P_0	0.1	MPa
Initial hydrogen concentration	c_0	0	wt%
H ₂ supply pressure	P_{abs}	7	MPa
HTF inlet temperature	$T_{f,in}$	298.15	K
HTF inlet velocity	v_f	0.4	m/s
Ambient temperature	T_{amb}	298.15	K

4. Results and discussion

4.1. Model parameterization and validation

The kinetic and thermodynamic parameters of the alloy were determined through experiment, and the complete multi-physics model was then validated against experimental data from the large-scale MHR module. The key geometric dimensions of the reactor and the baseline operating conditions used as a reference for the parametric studies are summarized in Table 2.

Based on previous studies, the JMAK model with a reaction order of $n = 1$ was determined to be most suitable for describing the hydrogen absorption process of the alloy [44,45]. Fig. 3(a) demonstrates the excellent agreement between the experimental data and the model fits at a hydrogen pressure of 6 MPa across different temperatures. Fig.

3(b) shows a similarly excellent agreement at a constant temperature of 273.15 K under various pressures. Subsequently, from an Arrhenius analysis, the pre-exponential factor A and activation energy E_a were determined to be 1.022 s^{-1} and 9.350 kJ/mol , respectively. Then, the kinetic equation for the hydrogenation process in terms of mass concentration was established.

$$\frac{\partial c}{\partial t} = 1.022 \exp\left(-\frac{9350}{RT}\right) \ln\left(\frac{P}{P_{eq}}\right) (c_{max} - c) \quad (26)$$

Fig. 3(c) shows the measured PCT isotherms. From these data, the Van't Hoff plot was constructed in Fig. 3(d), exhibiting a strong linear relationship ($R^2 = 0.9999$). From its slope and intercept, the enthalpy (ΔH) and entropy (ΔS) of absorption were determined to be 22.25 kJ/mol and 115.67 J/(mol K) . A continuous three-dimensional surface of equilibrium pressure $P_{eq}(c, T)$ was then constructed based on the reference isotherm at 293.15 K and the calculated thermodynamic parameters, as visualized in Fig. 3(e). This surface function was subsequently implemented in the numerical model. The model was validated by comparing its predictions against a dedicated experiment on the MHR module. The simulation replicated an absorption experiment with an initial uniform temperature of 298.15 K and a hydrogen supply pressure of 7 MPa, while the HTF was circulated through both cooling circuits at an inlet temperature of 298.15 K of the velocity of 0.4 m/s. Fig. 3(f) presents a comparison between the simulated temperature profiles (solid lines) and the experimentally measured data (symbols) at the four thermocouple locations at TC1–TC4. The model demonstrates excellent agreement with the experiment, accurately capturing the initial rapid temperature rise, the magnitude and timing of the peak temperatures, and the subsequent slow cooling phase. The Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) were calculated for all thermocouple locations. The overall average MAE and RMSE were found to be 2.25 K and 2.56 K, respectively, indicating a high degree of predictive accuracy. The proximity of the RMSE to the MAE values further confirms that no significant outlier deviations were observed. A detailed description of the error analysis methodology and results is provided in the Supplementary Material S6. Minor discrepancies can be attributed to model idealizations and potential experimental uncertainties. Overall, the strong correlation validates the predictive capability of the developed numerical model, confirming that it accurately represents the coupled thermo-kinetic behavior within the large-scale MHR. Furthermore, the robustness of the model's predictions was confirmed through a sensitivity analysis, with details provided in the Supplementary Material S7. The analysis shows that a $\pm 10\%$ variation in key uncertain parameters, such as the bed porosity (δ) and the pre-exponential factor (A), resulted in only a minor change in the peak average bed temperature ($< 0.6\%$) and a corresponding change in the 90% absorption time ranging from -6.7% to $+3.3\%$. This indicates that the study's main conclusions are robust and not artifacts of minor parameter uncertainties.

4.2. Effect of HTF velocity on MHR performance during hydrogen absorption

The velocity of the HTF is a critical operational parameter that directly dictates the rate of heat removal from the MHR. To quantify its impact on performance, a series of simulations were conducted for the hydrogen absorption process, varying the HTF inlet velocity (v_f) from 0.0 m/s to 0.8 m/s. The initial system temperature was set to 298.15 K, and the hydrogen supply pressure was 7 MPa.

Fig. 4 presents the time evolution of the volume-averaged MH bed temperature and reaction fraction for the different HTF velocities. As shown in Fig. 4(a), the temperature profiles exhibit a characteristic behavior driven by the exothermic absorption reaction. In all cases, the bed temperature initially rises rapidly as the reaction begins, reaching a peak before gradually declining as the reaction rate slows and heat is dissipated. The influence of the HTF velocity is pronounced. In the absence of active cooling ($v_f = 0.0 \text{ m/s}$), significant heat accumulation

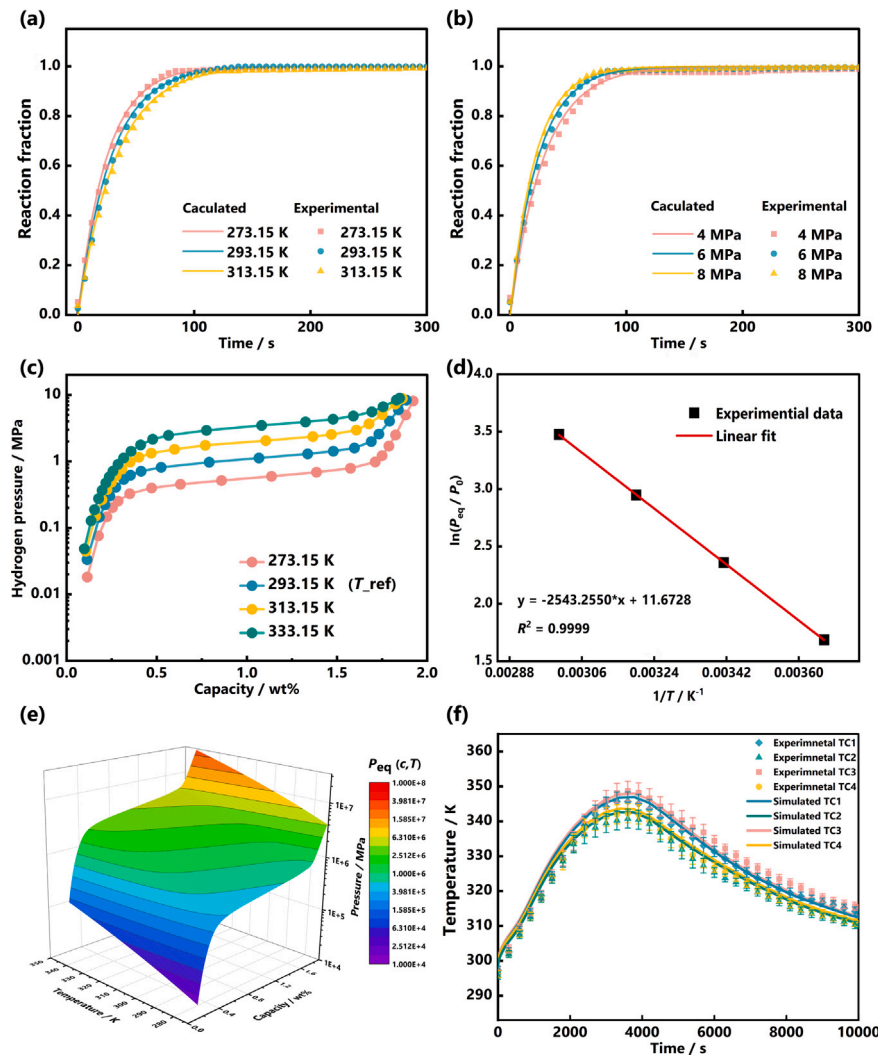


Fig. 3. (a) Comparison of experimental data and JMAK model fits for hydrogen absorption kinetics at a constant pressure of 6 MPa, where symbols represent the experimental data and solid lines are the model fits. (b) Comparison of experimental data and JMAK model fits for absorption kinetics at a constant temperature of 273.15 K, where symbols represent the experimental data and solid lines are the model fits. (c) Experimentally measured PCT isotherms for the alloy, where symbols are the measured equilibrium points. (d) Van't Hoff plot derived from the PCT data, where symbols are the derived data points and the solid line is the linear fit. (e) 3D surface plot of the calculated equilibrium pressure as a function of temperature and hydrogen concentration, where the color scale indicates the equilibrium pressure. (f) Comparison of simulated temperature profiles and experimental measurements within the large-scale MHR, where symbols represent the experimental measurements and solid lines are the corresponding simulated profiles.

occurs, leading to a peak average temperature of approximately 351.57 K. As the HTF velocity increases, the enhanced convective heat transfer leads to a substantial reduction in the peak temperature, which drops to approximately 337.04 K at $v_f = 0.8$ m/s. This demonstrates the effectiveness of the dual-circuit HX in managing the thermal load. The thermal behavior shown in Fig. 4(a) has a direct consequence on the hydrogen absorption kinetics, as illustrated in Fig. 4(b). The reaction fraction, defined as the ratio of the current hydrogen concentration to the maximum capacity ($c/c_{\max}$), is strongly dependent on the HTF velocity. The case with no active cooling exhibits the slowest absorption rate, taking over 10,000 s to reach approximately 86% completion. This is because the high bed temperature raises the local equilibrium pressure P_{eq} , significantly reducing the reaction driving force $\ln(P/P_{eq})$. Conversely, increasing the HTF velocity accelerates the absorption process dramatically. At $v_f = 0.8$ m/s, the reactor reaches over 90% capacity ($t_{90\%}$, is defined as the time at which the volume-averaged reaction fraction reaches a value of 0.9) in less than 5000 s. This accelerated performance is a direct result of the efficient heat removal, which keeps the bed temperature lower, thereby maintaining a large driving force for the reaction throughout the process.

The spatio-temporal distributions of temperature and reaction fraction are then examined. Fig. 5 displays the 3D temperature contours within the MHR at different time intervals. The images clearly visualize the “hot core” phenomenon characteristic of MH beds with poor thermal conductivity. Heat is generated throughout the bed but can only be effectively removed at the boundaries of the inner tube and the outer shell. In the $v_f = 0.0$ m/s case, the entire reactor core remains at a very high temperature for an extended period, severely hindering the reaction. As v_f increases, the effectiveness of the HXs becomes apparent. The temperature near the actively cooled surfaces is kept low, and the overall temperature field becomes significantly more uniform, with the size and intensity of the hot core being drastically reduced.

This temperature distribution directly governs the propagation of the reaction front, as shown by the 2D cross-sectional plots of the reaction fraction in Fig. 6. The reaction initiates at the coldest regions – the surfaces of the inner tube/fins and the outer shell – and propagates inwards. For the no-flow case, the reaction front advances very slowly after an initial absorption phase. Consequently, even after 5000 s, a significant portion of the core region exhibits a very low reaction fraction, indicating that the reaction has been severely inhibited by

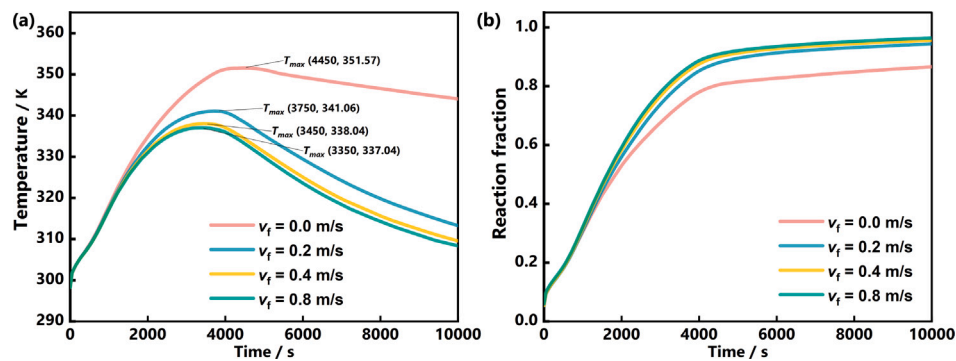


Fig. 4. Effect of HTF velocity on the overall MHR performance during hydrogen absorption for velocities ranging from 0.0 to 0.8 m/s over a 10 000 s period: (a) Averaged temperature of the MH bed versus time; (b) Averaged reaction fraction (normalized capacity, c/c_{max}) versus time. Constant parameters: $P_{abs} = 7$ MPa, $T_f = 298.15$ K.

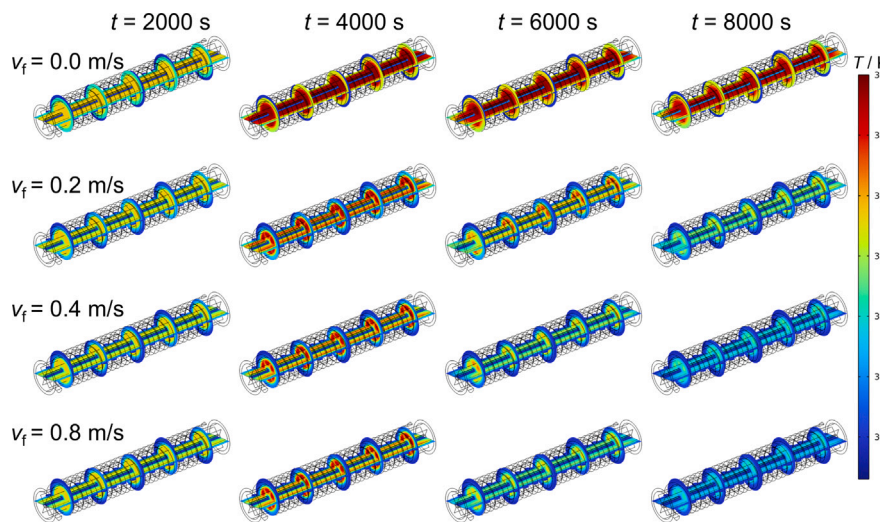


Fig. 5. 3D temperature distribution contours within the MHR at selected time intervals (2000, 4000, 6000, and 8000 s) for different HTF inlet velocities from 0.0 to 0.8 m/s. The color legend represents the local temperature (T/K). Constant parameters: $P_{abs} = 7$ MPa, $T_f = 298.15$ K.

high temperatures. As the HTF velocity increases, the reaction front moves much more rapidly and uniformly. At $v_f = 0.8$ m/s, the reaction has propagated deep into the bed by $t = 5000$ s, with the majority of the cross-section reaching a high state of reaction, although a small, less-reacted zone persists in the centermost areas. This visualization confirms that the performance of the MHR is not limited by the intrinsic material kinetics but by the rate at which reaction heat can be transported from the bed's interior to the actively cooled surfaces.

Furthermore, a critical trend of diminishing returns with increasing HTF velocity is evident from the results. While the performance improvement from $v_f = 0.0$ m/s to 0.4 m/s is substantial, the additional gain from increasing the velocity further to 0.8 m/s is marginal. This is clearly observed in Fig. 4, where the temperature and reaction fraction curves for 0.4 m/s and 0.8 m/s are closely clustered, indicating a much smaller performance gap compared to the lower velocity ranges. The spatial distributions in Fig. 5 and Fig. 6 corroborate this finding; the temperature and reaction fraction contours for 0.4 m/s and 0.8 m/s are remarkably similar, especially at later times, suggesting that the internal state of the reactor is no longer significantly improved by the faster HTF flow. The overall heat transfer process from the bed's core to the HTF is governed by two primary thermal resistances in series: the convective resistance within the thermal boundary layers of the HTF and the conductive resistance within the MH bed itself. Increasing the HTF velocity primarily reduces the convective resistance. At lower flow rates, this reduction significantly lowers the total thermal resistance, leading to substantial performance gains. However, as

the HTF velocity becomes sufficiently high, the convective resistance becomes a less significant fraction of the total resistance, which is now increasingly dominated by the large, intrinsic thermal resistance of the low-conductivity MH powder. Consequently, further increases in HTF velocity yield only minor improvements in overall heat transfer and, therefore, in reaction performance. This suggests the existence of an optimal HTF flow rate, beyond which increasing pumping power does not provide a proportional benefit in hydrogen absorption rates. This trade-off is quantified by an energy cost analysis, which shows that increasing the HTF velocity from 0.4 m/s to 0.8 m/s causes the required pumping power to rise nearly sevenfold from 0.23 W to 1.50 W, yet this significant energy penalty only reduces the 90% absorption time by approximately 6%, from 4500 s down to 4200 s (see Supplementary Material S8). Furthermore, a comparative simulation reveals that switching to a counter-current flow configuration offers no significant performance advantage, reinforcing the conclusion that the system is predominantly limited by internal conduction rather than the specifics of the convective heat exchange (see Supplementary Material S9).

4.3. Effect of HTF circuit configuration on MHR performance

The performance of different cooling circuit configurations was quantitatively analyzed to dissect the contributions of the dual-circuit system and its potential synergistic effects. Two primary single-circuit cases were considered: active outer cooling only ($v_i = 0, v_o = 0.4$ m/s)

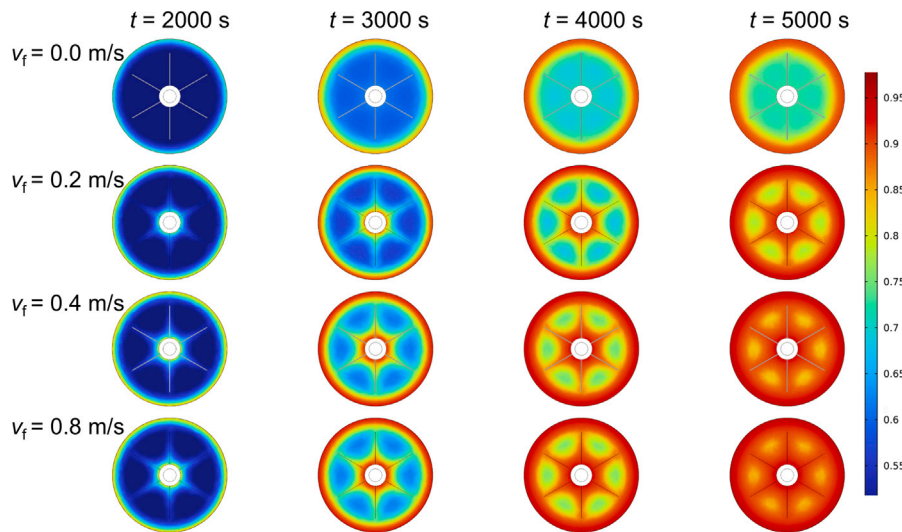


Fig. 6. 2D cross-sectional distribution of the reaction fraction within the MH bed at selected time intervals (2000, 3000, 4000, and 5000 s) for different HTF inlet velocities from 0.0 to 0.8 m/s. The color legend represents the local reaction fraction $(c/c_{\max})$. Constant parameters: $P_{\text{abs}} = 7$ MPa, $T_f = 298.15$ K.

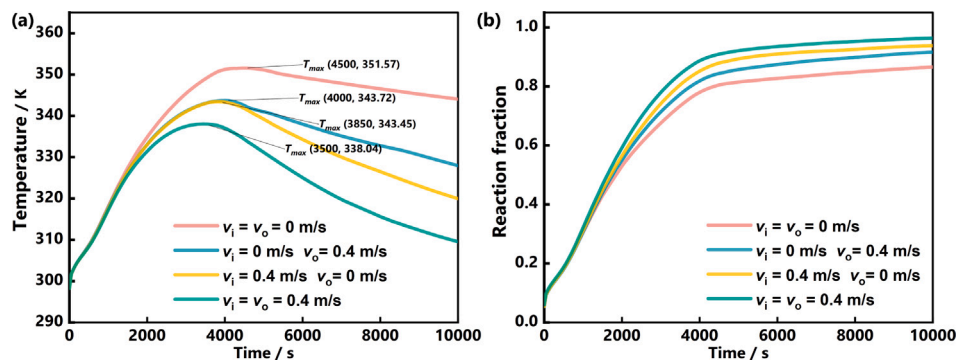


Fig. 7. Effect of different cooling strategies (no cooling, outer only, inner only, and dual cooling) on the overall MHR performance during hydrogen absorption: (a) Averaged temperature of the MH bed versus time; (b) Averaged reaction fraction $(c/c_{\max})$ versus time. For single-circuit cases, the active velocity is 0.4 m/s. Constant parameters: $P_{\text{abs}} = 7$ MPa, $T_f = 298.15$ K.

and active internal cooling only ($v_i = 0.4, v_o = 0$ m/s). As shown in Fig. 7(a), with only passive natural convection, the bed temperature peaks at approximately 351.57 K. Activating either the external or internal cooling circuit individually provides a significant performance enhancement, suppressing the peak temperature to approximately 343 K. The internal cooling strategy proves slightly superior to the external one, as it offers a greater temperature reduction in the later stages, which corresponds to the faster absorption rate shown in Fig. 7(b). When both circuits are active simultaneously, the system exhibits the best performance, with the peak temperature suppressed to approximately 338.04 K and the fastest overall absorption rate.

Fig. 8 details the temperature distribution under these different cooling strategies. With outer cooling only, effective heat removal is confined to the reactor's periphery, leaving the broad central region, particularly the powder areas between the fins, at high temperatures. In contrast, with internal cooling only, the central tube and its radial fins create an effective heat path from the core outwards, displacing the highest temperature zones to the powder regions located midway between the inner fins and the outer wall. The superiority of the dual cooling strategy is evident, as it creates the most uniform temperature field by actively cooling both the central finned structure and the outer boundary simultaneously. This temperature distribution directly governs the hydrogen absorption progress, as summarized in Fig. 9, where the reaction front consistently advances most rapidly along the effectively cooled surfaces.

To test whether the benefit of the dual-circuit system is more than just the effect of a higher total flow rate, single-circuit simulations were subsequently performed under conditions matching the total mass flow rate of the baseline dual-circuit case ($v_i = 0.4, v_o = 0.4$ m/s). The operating velocities for these single-circuit cases were calculated accordingly (see Supplementary Material S10 for calculations). As shown in Figure S3 in the Supplementary Material, even at the equivalent total mass flow rate, the average bed temperature of the single-circuit cases is significantly higher than that of the dual-circuit case, while their reaction rates are lower.

A summary of key performance metrics for all operating cases is presented in Table 3. A trend of diminishing returns with increasing flow rate is observed: tripling the external circuit's flow rate shortens $t_{90\%}$ by only 12%, while a 50% increase in the internal circuit's flow rate yields a 3% reduction. More importantly, the baseline dual-circuit configuration ($t_{90\%} = 4500$ s) is significantly faster than the single-circuit configurations operating at the equivalent total flow rate (5190 s and 7140 s). The dual-circuit system is 13.3% faster than even the more effective inner circuit operating alone at the higher equivalent flow. This result demonstrates that distributing the cooling resource to manage both the core and periphery simultaneously is more effective than concentrating it in a single location, thus confirming a synergistic effect between the two cooling circuits. The mechanism for this synergy is that the reaction front in the dual-circuit system can

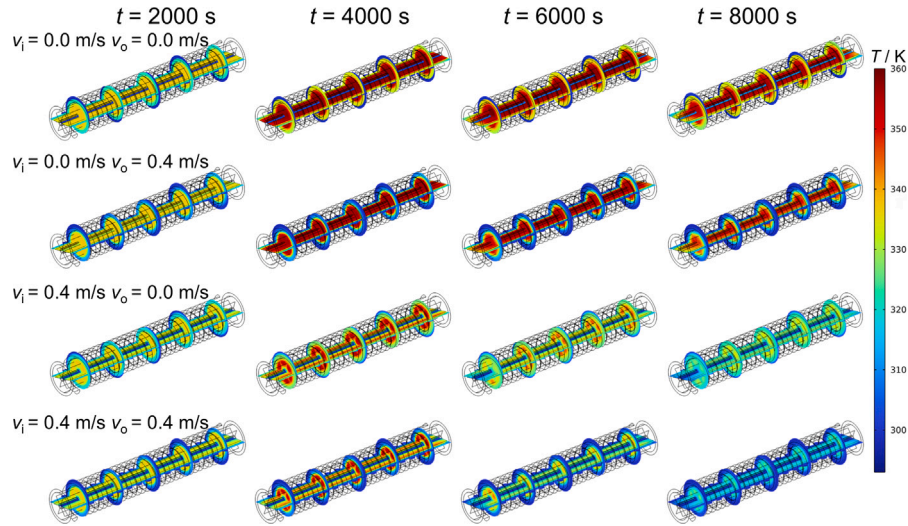


Fig. 8. 3D temperature distribution contours within the MHR at selected time intervals for different cooling strategies. The color legend represents the local temperature (T/K). Constant parameters: $P_{abs} = 7$ MPa, $T_f = 298.15$ K.

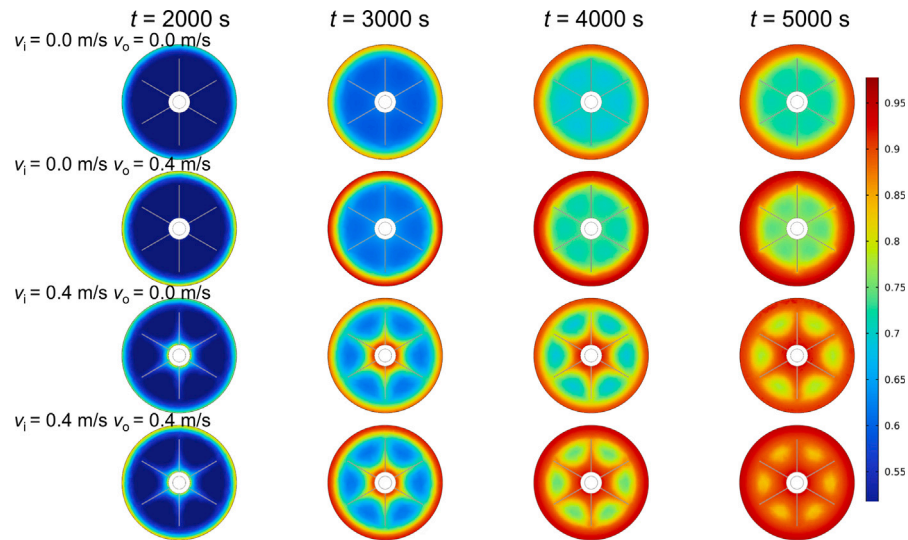


Fig. 9. 2D cross-sectional distribution of the reaction fraction within the MH bed at selected time intervals for different cooling strategies. The color legend represents the local reaction fraction (c/c_{max}). Constant parameters: $P_{abs} = 7$ MPa, $T_f = 298.15$ K. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

advance efficiently from both the inner and outer boundaries. This superior thermal uniformity maintains a high thermodynamic driving force across a larger volume of the bed, resulting in the most efficient and complete utilization of the hydride material.

4.4. Effect of solid-phase thermal conductivity on MHR performance during hydrogen absorption

The intrinsic thermal conductivity of the metal hydride solid phase λ_s is a key material property governing internal heat diffusion. To quantify its impact on module performance, we varied λ_s from 1.6 to 9.6 W/(m K) - a range representing materials from pure powders to thermally-enhanced composites. Common thermal enhancement methods include incorporating high-conductivity additives like expanded natural graphite or compacting the alloy powder with binders to minimize inter-particle contact resistance. For instance, a recent study demonstrated that a novel compaction process could enhance the effective thermal conductivity of a Ti-based alloy bed to over 10 W/(m K),

Table 3

Quantitative comparison of different cooling strategies under standard and equivalent total flow rate conditions.

Cooling strategy	Inner velocity [m/s]	Outer velocity [m/s]	$t_{90\%}$ [s]	$T_{peak,avg}$ [K]
A. Standard flow rate comparison ($v = 0.4$ m/s)				
No cooling	0	0	>10000	351.57
Outer cooling only	0	0.4	8120	343.72
Inner cooling only	0.4	0	5350	343.45
Dual cooling	0.4	0.4	4500	338.04
B. Synergy analysis (Equivalent total mass flow rate)				
Outer cooling only	0	1.20	7140	343.46
Inner cooling only	0.60	0	5190	342.71

a dramatic improvement over loose powder [46]. While practical challenges such as reduced storage density or long-term cycling stability exist, these advanced techniques confirm that achieving a conductivity in the 5–10 W/(m K) range is viable for industrial applications, thus

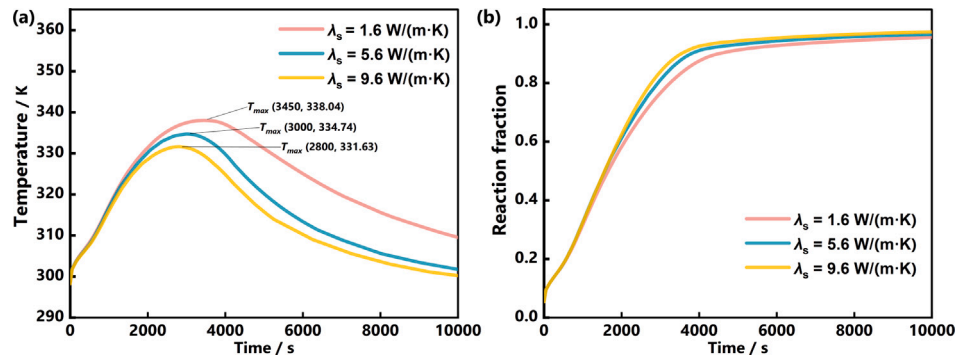


Fig. 10. Effect of solid-phase thermal conductivity (λ_s) on the overall MHR performance during hydrogen absorption for λ_s values from 1.6 to 9.6 W/(m K): (a) Averaged temperature of the MH bed versus time; (b) Averaged reaction fraction (c/c_{max}) versus time. Constant parameters: $\nu_f = 0.4$ m/s, $P_{abs} = 7$ MPa, $T_{initial} = 298.15$ K.

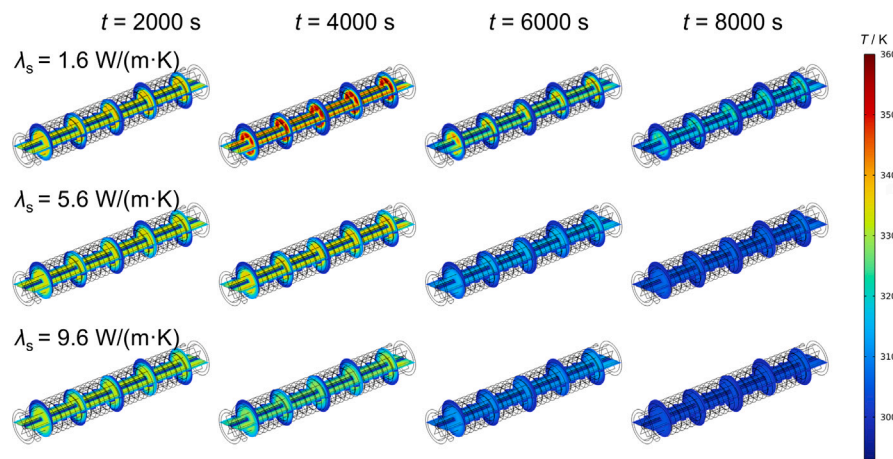


Fig. 11. 3D temperature distribution contours within the MHR at selected time intervals for different solid-phase thermal conductivities (λ_s) from 1.6 to 9.6 W/(m K). The color legend represents the local temperature (T/K). Constant parameters: $\nu_f = 0.4$ m/s, $P_{abs} = 7$ MPa, $T_{initial} = 298.15$ K. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

validating the relevance of the high-conductivity scenarios explored in our work. Other parameters were held constant: an HTF velocity of 0.4 m/s, an initial temperature of 298.15 K, and a supply pressure of 7 MPa.

As shown in Fig. 10, increasing the solid-phase thermal conductivity directly leads to improved performance. A higher λ_s value results in a lower peak bed temperature in Fig. 10(a) and consequently, a significantly faster hydrogen absorption rate in Fig. 10(b). For instance, increasing λ_s from 1.6 to 9.6 W/(m K) reduces the peak average bed temperature by approximately 7 K, from 338.04 K to 333.11 K and reduces the time required to reach 90% capacity by approximately 24%, from 4500 s to 3400 s.

The primary mechanism for this enhancement is the promotion of thermal uniformity within the reactor. Figs. 11 and 12 provide a clear visualization of this effect. At low thermal conductivity $\lambda_s = 1.6$ W/(m K), heat generated in the reactor's core becomes trapped, creating a large temperature gradient between the hot center and the actively cooled peripheries as shown in Fig. 11. This “hot core” phenomenon severely inhibits the local reaction rate, causing the reaction front to stall, as seen in the non-uniform reaction fraction distribution as shown in Fig. 12. Conversely, a high thermal conductivity $\lambda_s = 9.6$ W/(m K) creates an effective “thermal bridge” between the core and the cooling surfaces. This dramatically reduces internal temperature gradients as shown in Fig. 11 leading to a much more isothermal reaction environment. As a result, the reaction front propagates more

uniformly throughout the bed, allowing the entire volume of the MH material to react more simultaneously and efficiently as shown in Fig. 12.

4.5. Effect of inlet HTF temperature on MHR performance during hydrogen absorption

The inlet temperature of the HTF, T_f , sets the thermal boundary for the reactor and its effect was investigated by simulating absorption with HTF inlet temperatures of 278.15 K, 288.15 K, and 298.15 K, while other parameters were held constant with HTF velocity = 0.4 m/s, $\lambda_s = 1.6$ W/(m K), and a supply pressure for absorption $P_{abs} = 7$ MPa.

As shown in Fig. 13, a lower HTF temperature consistently leads to superior absorption performance. The case with the lowest HTF temperature at $T_f = 278.15$ K exhibits both the low bed temperature in Fig. 13(a) and the fastest overall hydrogen absorption rate in Fig. 13(b). This result occurs because temperature has two opposing influences on this exothermic reaction. While lower temperatures slow the chemical reaction rate as described by the Arrhenius relationship $\exp(-E_a/RT)$, they also significantly increase the thermodynamic driving force by suppressing the equilibrium pressure P_{eq} . The data in Fig. 13(b) conclusively demonstrates that for this system, the enhancement of the thermodynamic driving force is the dominant mechanism, far outweighing any kinetic penalty and leading to a faster net absorption process.

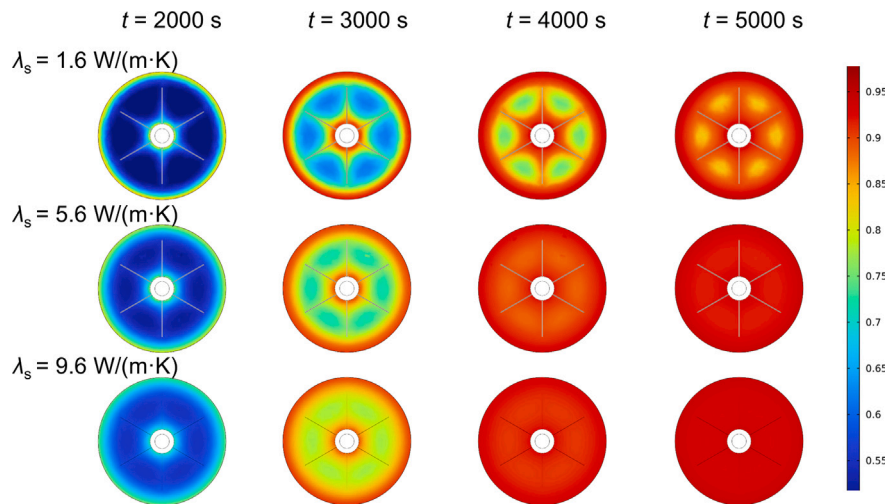


Fig. 12. 2D cross-sectional distribution of the reaction fraction within the MH bed at selected time intervals for different solid-phase thermal conductivities (λ_s) from 1.6 to 9.6 W/(m K). The color legend represents the local reaction fraction ($c/c_{\max}$). Constant parameters: $v_f = 0.4$ m/s, $P_{\text{abs}} = 7$ MPa, $T_f = 298.15$ K.

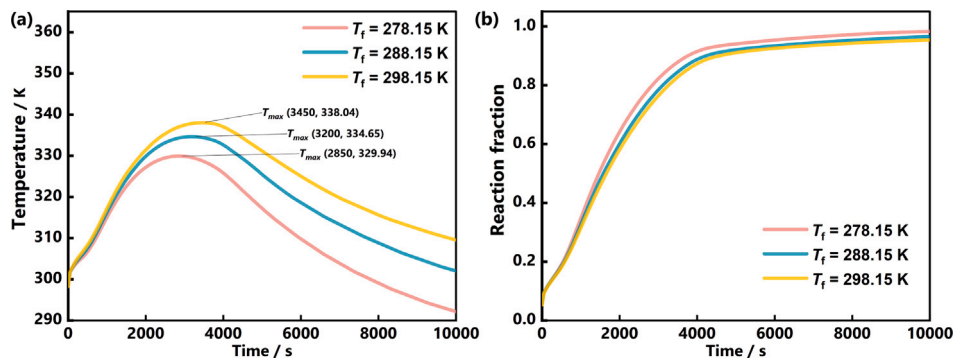


Fig. 13. Effect of HTF inlet temperature (T_f) on the overall MHR performance during hydrogen absorption for T_f values of 278.15 K, 288.15 K, and 298.15 K: (a) Averaged temperature of the MH bed versus time; (b) Averaged reaction fraction ($c/c_{\max}$) versus time. Constant parameters: $v_f = 0.4$ m/s, $\lambda_s = 1.6$ W/(m K), $P_{\text{abs}} = 7$ MPa.

Fig. 14 confirms that a lower HTF temperature results in a significantly cooler thermal environment throughout the entire bed at all times. As a direct consequence, the reaction front propagation, shown in Fig. 15, is much more efficient at lower HTF temperatures. In the $T_f = 278.15$ K case, the reaction front advances swiftly and uniformly, leading to a nearly complete reaction across the entire cross-section by $t = 5000$ s. In contrast, at higher HTF temperatures, the reaction slows prematurely in the warmer core regions, resulting in a less complete and less uniform absorption profile at the same point in time.

4.6. Effect of supply pressure on MHR performance during hydrogen absorption

The supply pressure for absorption P_{abs} was varied between 5, 7, and 9 MPa to investigate its effect on the MHR's performance, with other conditions held constant: an HTF inlet temperature of 298.15 K at 0.4 m/s, and a solid-phase thermal conductivity $\lambda_s = 1.6$ W/(m K).

The averaged temperature of the MH bed is shown in Fig. 16(a). A higher target pressure results in a higher peak average temperature, increasing from 333.15 K at 5 MPa to 340.84 K at 9 MPa. The time to reach this peak is also shorter at higher pressures. After peaking, the 9 MPa case exhibits the fastest cooling rate, and its temperature drops below the other cases after approximately 5000 s. This behavior is a direct consequence of the absorption kinetics, presented in Fig. 16(b), which show that a higher target pressure leads to a significantly faster reaction. Increasing the supply pressure from 5 to 9 MPa reduces the

time required to reach 90% of the total storage capacity by more than half, from 7500 s down to just 3600 s. At 9 MPa, the near-completion of the reaction causes the internal heat generation to cease much earlier, allowing for the subsequent rapid cooling phase.

The spatial temperature distribution within the bed is presented in Fig. 17. At higher pressures, the initial temperature distribution is less uniform, with steeper thermal gradients due to the more intense and rapid heat generation. However, in the later stages at $t = 8000$ s, the 9 MPa case exhibits a more uniform and cooler temperature field, a result of its earlier reaction completion. This thermal evolution also governs the reaction front propagation shown in Fig. 18. A higher pressure drives a more aggressive and swift reaction front through the bed, consistent with the faster overall absorption rates observed. In summary, despite the more intense heat release at higher pressures, the overall absorption process is accelerated rather than inhibited. This indicates that for the range of pressures studied, the net effect of the enhanced thermodynamic driving force is dominant, effectively overcoming the negative feedback from the elevated bed temperature.

4.7. Comparative analysis with jacketed shell configuration systems

To contextualize the contributions of this work, our reactor is compared with other recently reported large-scale, jacketed shell type metal hydride systems. The present study investigates a reactor with a 400 kg alloy capacity per vessel, which is a significant scale-up compared to other recent experimental and numerical studies. For instance, Karmakar et al. [47] experimentally studied a 10 kg multi-tube

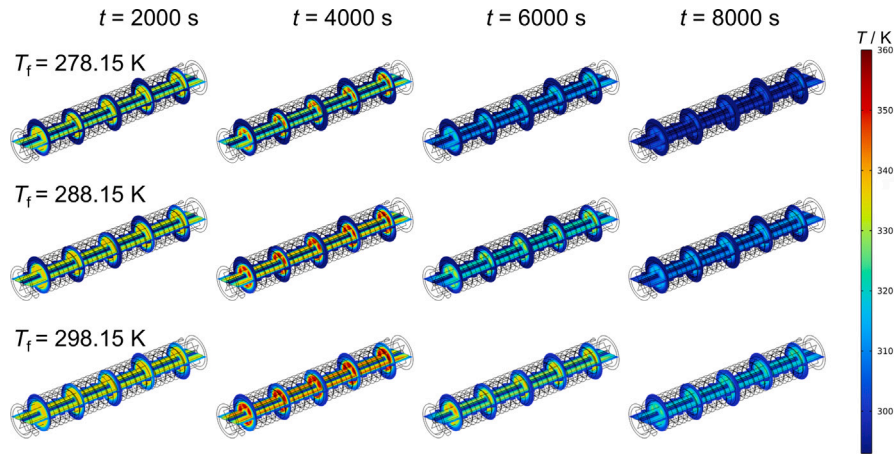


Fig. 14. 3D temperature distribution contours within the MHR at selected time intervals for different HTF inlet temperatures (T_f). The color legend represents the local temperature (T/K). Constant parameters: $v_f = 0.4$ m/s, $\lambda_s = 1.6$ W/(m K), $P_{abs} = 7$ MPa.

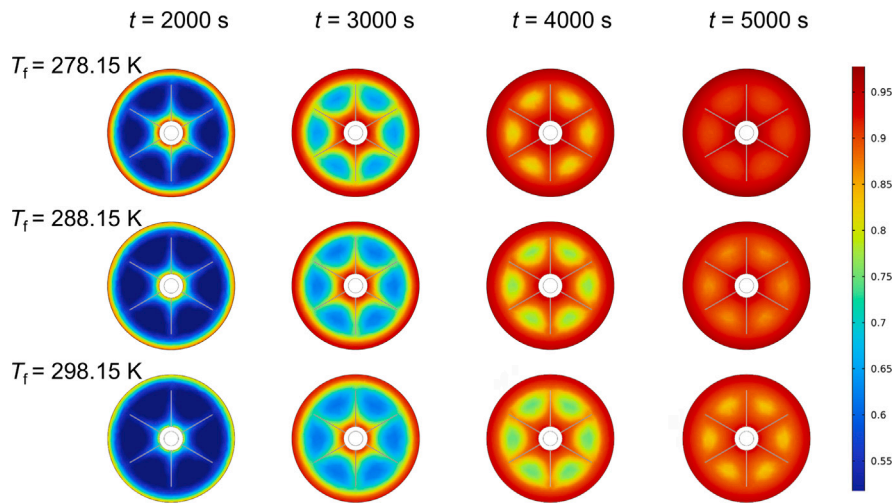


Fig. 15. 2D cross-sectional distribution of the reaction fraction within the MH bed at selected time intervals for different HTF inlet temperatures (T_f). The color legend represents the local reaction fraction (c/c_{max}). Constant parameters: $v_f = 0.4$ m/s, $\lambda_s = 1.6$ W/(m K), $P_{abs} = 7$ MPa.

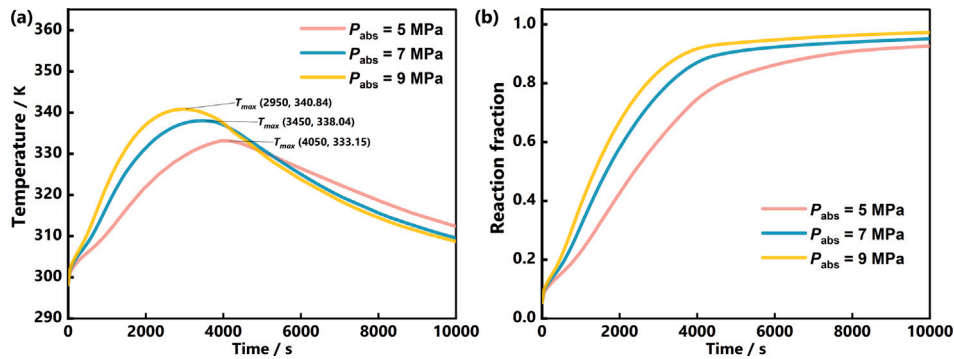


Fig. 16. Effect of supply pressure (P_{abs}) on the overall MHR performance during hydrogen absorption for P_{abs} values of 5, 7, and 9 MPa: (a) Averaged temperature of the MH bed versus time; (b) Averaged reaction fraction (c/c_{max}) versus time. Constant parameters: $v_f = 0.4$ m/s, $\lambda_s = 1.6$ W/(m K), $T_f = 298.15$ K.

reactor cooled by an external water jacket, Prasad & Muthukumar [31] and Parashar et al. [30] experimentally investigated the performance of a 9 kg shell-and-tube reactor, while Afzal et al. [29] numerically investigated the performance of a 50 kg shell-and-tube reactor. The increase in scale in our work from the common 10–50 kg class to the 400 kg class provides critical performance data under thermal management challenges that are far more severe than those in smaller

systems. Despite the 40-fold increase in alloy mass, the absorption time to reach 90% capacity in our system (approximately 3600 s at 9 MPa) remains remarkably competitive, being on a comparable timescale to the 680–1620 s reported for these much smaller 9–10 kg systems under their respective optimal conditions. Conventional shell-and-tube or jacketed designs often rely on high HTF flow rates to manage the reaction heat. The 10 kg experimental reactor of Karmakar et al. for

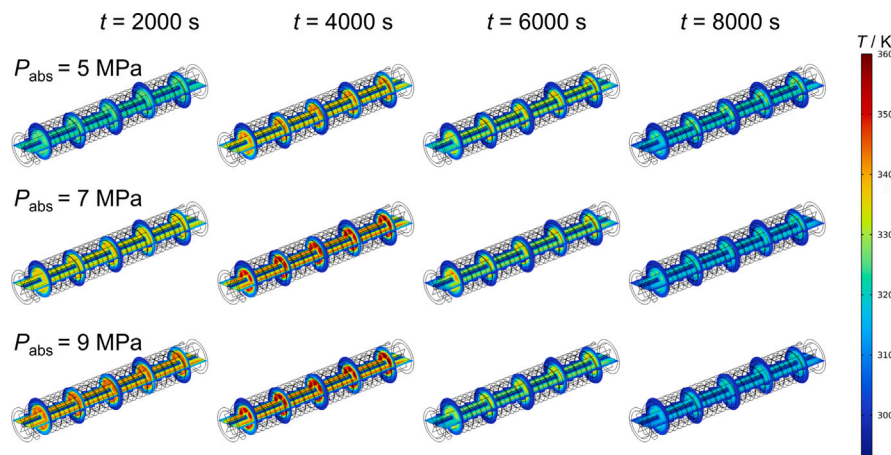


Fig. 17. 3D temperature distribution contours within the MHR at selected time intervals for different supply pressures (P_{abs}). The color legend represents the local temperature (T/K). Constant parameters: $v_f = 0.4$ m/s, $\lambda_s = 1.6$ W/(m K), $T_f = 298.15$ K.

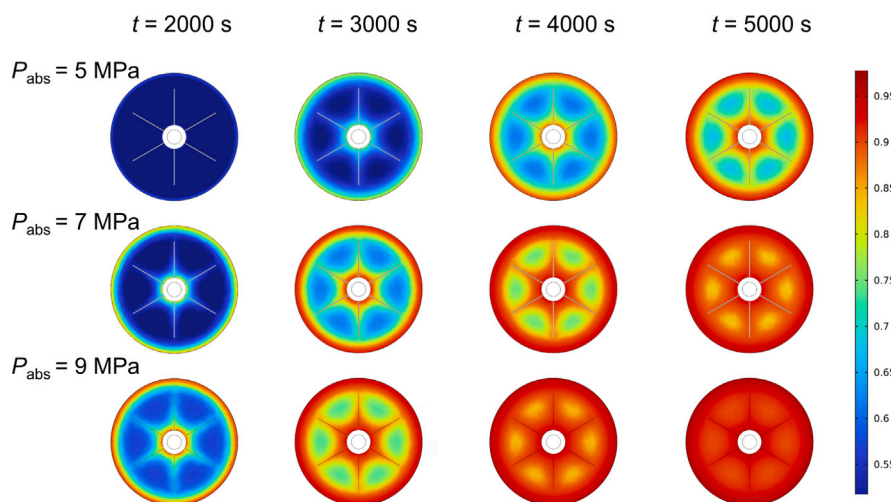


Fig. 18. 2D cross-sectional distribution of the reaction fraction within the MH bed at selected time intervals for different supply pressures (P_{abs}). The color legend represents the local reaction fraction (c/c_{max}). Constant parameters: $v_f = 0.4$ m/s, $\lambda_s = 1.6$ W/(m K), $T_f = 298.15$ K. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

example, required an HTF flow rate of 20 LPM, while simulations by Afzal et al. for their 50 kg shell-and-tube system explored a range of 30–90 LPM. However, their work also revealed that this approach yields diminishing returns, as performance saturates when the system becomes limited by the bed's internal conduction rather than external convection. In contrast, the dual-circuit helical coil design in the present study manages the thermal load of a 400 kg system using a total HTF flow rate of 25.45 LPM (at a reference velocity of 0.4 m/s), which is comparable to that of the 10 kg Karmakar et al. reactor. This highlights the superior heat transfer efficiency of the external helical coil compared to conventional jacketed shell configurations, providing a more effective and scalable thermal management solution.

5. Limitations and future work

This study provides a detailed thermo-kinetic analysis of a novel dual-circuit MHR, but it is subject to certain limitations that open avenues for future research. The current work focuses on the performance during a single absorption cycle under idealized boundary conditions and within a specific range of operating parameters, which was chosen to elucidate key physical phenomena such as the heat transfer bottleneck shift. For this technology to progress towards practical application, several key areas warrant further investigation. A

primary direction for future work is the study of long-term performance and reliability. The current model assumes constant material properties, whereas real-world applications involve thousands of cycles, leading to material degradation such as pulverization. This can alter the bed's thermophysical properties (e.g., thermal conductivity and permeability), impacting system performance over time. Future research should therefore involve long-term experimental cycling to quantify degradation, coupled with advanced modeling that incorporates these evolving properties to predict the system's full life-cycle performance. Furthermore, a deeper optimization of the thermal management system is required. This includes a systematic investigation of materials and methods to enhance the bed's thermal conductivity, along with a critical trade-off analysis between thermal gains and any loss in hydrogen storage density. Moreover, exploring alternative cooling strategies, such as the integration of heat pipes, phase change materials, or the use of advanced manufacturing techniques like 3D printing for bionic heat exchangers, could offer pathways to higher efficiency. Finally, the successful scaling and commercialization of this modular concept depend on addressing practical engineering challenges. A comprehensive techno-economic analysis is needed to evaluate manufacturing costs, and a rigorous safety assessment is crucial, particularly concerning the long-term integrity of the internal cooling circuit in a high-pressure hydrogen environment. Future modeling could also incorporate dynamic environmental conditions and explore the system's behavior at

very high pressures, where it transitions into a hybrid gas–solid storage regime, to fully map its operational envelope.

6. Conclusion

Through an experimentally validated numerical model, this study quantifies the thermo-kinetic performance of a large-scale, modular MHR equipped with a novel dual-circuit heat exchanger. The reactor's performance is co-limited by the external convection provided by the dual circuits and the bed's own internal conduction. While increasing the HTF velocity within the circuits from 0 to 0.8 m/s effectively reduces the peak average bed temperature by approximately 15.7 K, this benefit exhibits diminishing returns as the system bottleneck shifts to poor internal thermal conduction. Analysis of the cooling strategies further reveals that the internal cooling circuit is slightly more effective than the external one, while the synergistic operation of the dual circuits achieves the optimal heat transfer and fastest absorption rate by simultaneously cooling the core and periphery, confirming the advantage of this hybrid design. Consequently, enhancing the bed's solid-phase thermal conductivity is crucial, with an increase from 1.6 to 9.6 W/(m K) shortening the time to reach 90% capacity by approximately 24%. Fundamentally, the entire absorption process is governed by the thermodynamic driving force rather than kinetics, as demonstrated by the fact that both lower HTF temperatures and higher supply pressures accelerate absorption. Increasing the supply pressure from 5 to 9 MPa reduces the time required to reach 90% of the total storage capacity by more than half, from 7500 s down to just 3600 s. The proposed reactor also manages the thermal load of a 400 kg alloy bed with an HTF flow rate (25.45 LPM) comparable to that used in 10 kg-class conventional jacketed shell configurations. These findings establish a quantitative framework for co-optimizing system components, providing guidance for enhancing the performance of future large-scale solid-state hydrogen storage systems.

Nomenclature

Abbreviations

BDF	Backward Differentiation Formula
FEM	Finite Element Method
HTF	Heat Transfer Fluid
HX	Heat Exchanger
JMAK	Johnson–Mehl–Avrami–Kolmogorov
LTE	Local Thermal Equilibrium
MH	Metal Hydride
MHR	Metal Hydride Reactor
PCT	Pressure–Composition–Temperature
RANS	Reynolds-Averaged Navier–Stokes
TC	Thermocouple

Symbols

A	Pre-exponential factor (s^{-1})
c	Hydrogen concentration (wt%)
c_{max}	Maximum hydrogen storage capacity (wt%)
C_p	Specific heat capacity ($J\ kg^{-1}\ K^{-1}$)
d_p	Mean particle diameter (m)
d_s	Fin thickness (m)
E_a	Apparent activation energy ($J\ mol^{-1}$)
H_e	External convective heat transfer coefficient ($W\ m^{-2}\ K^{-1}$)
I	Identity matrix
k	Turbulent kinetic energy ($m^2\ s^{-2}$)

K	Permeability of the MH bed (m^2)
M_{H_2}	Molar mass of hydrogen ($kg\ mol^{-1}$)
$\mathbf{n}$	Outward normal vector
P	Pressure (Pa)
p_f	Pressure of fluid (HTF) (Pa)
P_{iso}	Reference isotherm pressure (Pa)
Pr_T	Turbulent Prandtl number
q_{net}	Net heat flux per unit area ($W\ m^{-2}$)
$\mathbf{q}_t$	Tangential heat flux vector ($W\ m^{-1}$)
Q_m	Mass sink term ($kg\ m^{-3}\ s^{-1}$)
Q_T	Volumetric heat source ($W\ m^{-3}$)
R	Universal gas constant ($J\ mol^{-1}\ K^{-1}$)
R^2	Coefficient of determination
t	Time (s)
$t_{90\%}$	Time to reach 90% absorption (s)
T	Temperature (K)
$\mathbf{u}$	Velocity vector ($m\ s^{-1}$)
v	Scalar speed ($m\ s^{-1}$)

Greek symbols

δ	Porosity of the MH bed
ΔH	Enthalpy of absorption ($J\ mol^{-1}$)
ΔS	Entropy of absorption ($J\ mol^{-1}\ K^{-1}$)
ϵ	Turbulent dissipation rate ($m^2\ s^{-3}$)
Γ	Boundary
λ	Thermal conductivity ($W\ m^{-1}\ K^{-1}$)
μ	Dynamic viscosity (Pa s)
∇	Nabla operator
∇_t	Surface gradient operator
ρ	Density ($kg\ m^{-3}$)
τ	Stress tensor (Pa)
Φ_v	Viscous dissipation ($W\ m^{-3}$)

Subscripts

0	Initial state
abs	Absorption
amb	Ambient
atm	Atmospheric
e	Effective property
eq	Equilibrium
f	Fluid (HTF)
fin	Fin
g	Gas
i	Inner
in	Inlet
max	Maximum
o	Outer
ramp	Ramp
ref	Reference
s	Solid
T	Turbulent
wall	Wall

CRedit authorship contribution statement

Liqiang Ji: Writing – original draft, Data curation. **Dong Zhang:** Writing – review & editing, Funding acquisition, Conceptualization. **Wei Li:** Investigation, Formal analysis. **Jia Chen:** Writing – review & editing, Formal analysis. **Xitao Wang:** Validation, Supervision, Conceptualization. **Baozhong Liu:** Project administration, Funding acquisition. **Wei Han:** Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge the financial support for this research from the National Key Research and Development Program of China (Grant No. 2022YFB3803800), the National Natural Science Foundation of China (Grant No. U22A20120), the Fundamental Research Funds for the Universities of Henan Province, China (Grant No. NSFRF2502043), the Postdoctoral Research Fund of Henan Polytechnic University, China (Grant No. 712108/495), and the Doctoral Fund of Henan Polytechnic University, China (Grant No. 762407/048).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ijheatmasstransfer.2025.127987>.

Data availability

Data will be made available on request.

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